INCH-POUND

MIL-DTL-25427/7
08 December 2009

DETAIL SPECIFICATION SHEET

COUPLING FLANGE, AIRCRAFT MOUNTING, HYDRAULIC, SELF-SEALING

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-25427.

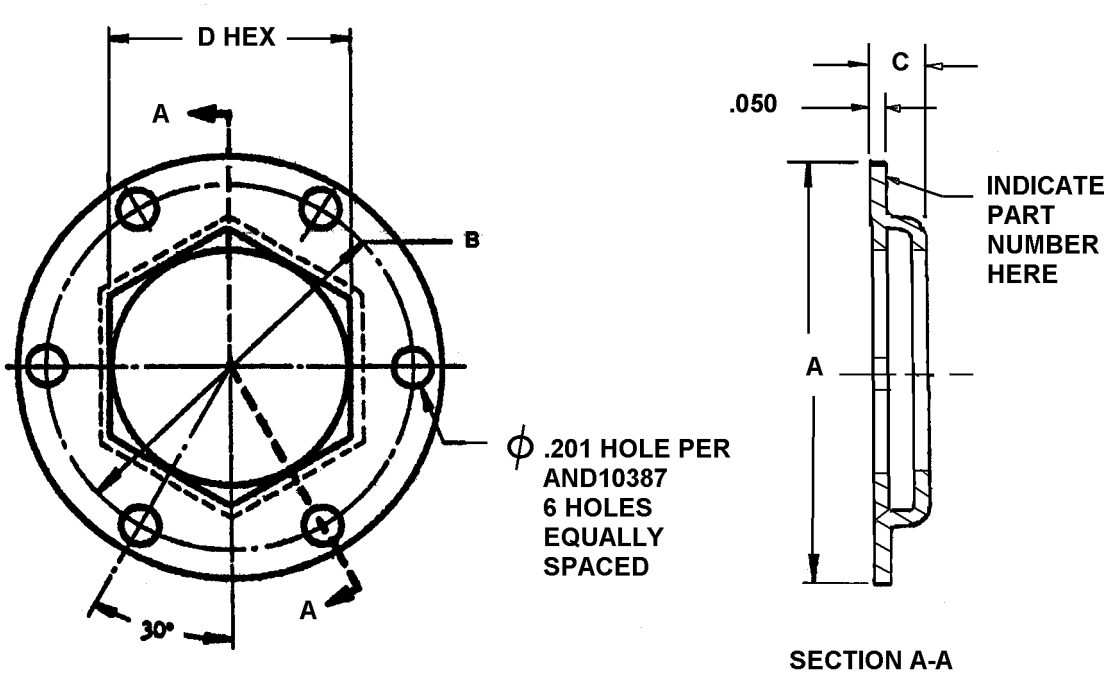

FIGURE 1. Flange coupling, dimensions, and configuration.

AMSC N/A FSC 4730

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TABLE I. Flange coupling dimensions.

Dash No.	Tube OD	A		B ± .005	C		D	
-4	0.250	1.812	± .010	1.440	0.189	+.006 -.007	1.070	+.020 -.007
-6	0.375	2.005		1.690	0.224		1.140	±.010
-8	0.50	2.005		1.690	0.224		1.240	
-10	0.625	2.500		2.120	0.224		1.640	
-12	0.75	2.500		2.120	0.230		1.640	
-16	1.00	2.750		2.380	0.224		1.890	
-20	1.25	3.000	± .015	2.620	0.225	+.005 -.008	2.145	± .005
-24	1.50	3.250		2.880	0.226	+.006 -.007	2.400	± .010

NOTES:

1. Dimensions are in inches, unless otherwise specified, tolerances: decimals ± .010.

2. Break all sharp edges and remove all hanging burrs and slivers.

3. For design feature purposes, this specification sheet takes precedence over procurement document.

4. Referenced documents shall be of the issue in effect on date of contract award.

REQUIREMENTS:

1. Material -Code S 300 series stainless steel or PH stainless steel or Code C carbon steel (see MIL-DTL-25427 for detail material specifications).

2. Finish -for finish requirements of steel, see MIL-DTL-25427.

3. Marking shall consist of the part number, manufacturer's name, trademark or CAGE code and shall be impression stamped or laser etched. Marking shall be protected from corrosion.

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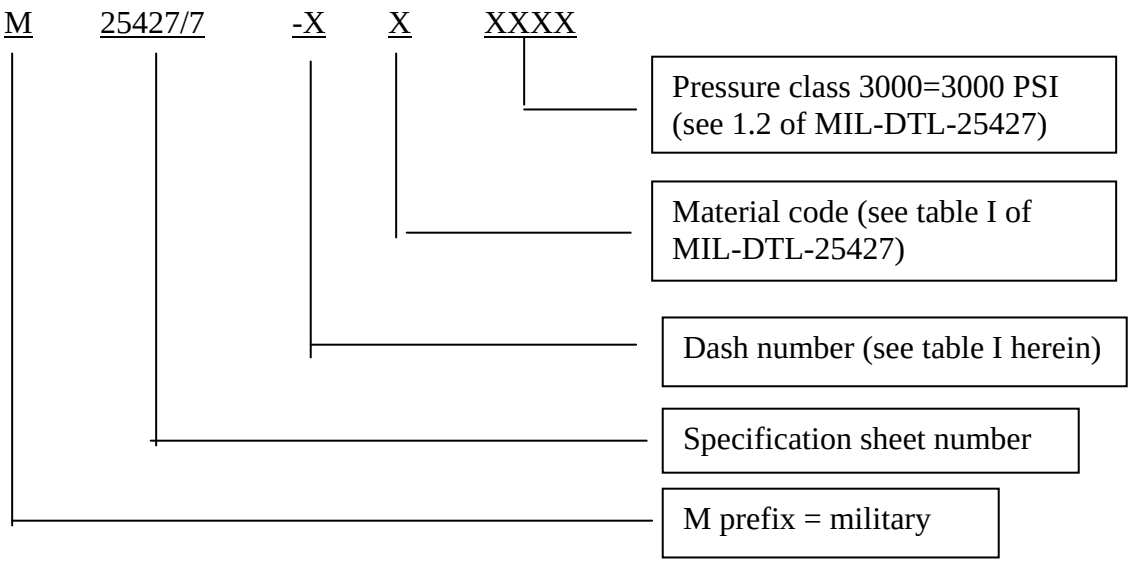
4. Part or identification number (PIN). The PIN consists of the letter M, the specification sheet number, a dash number from table I, material code, and pressure class.



Example of part number: M25427/7-8C5000 represents 0.5 tube OD carbon steel coupling flange, 5000 PSI operating pressure

5. Test requirements. The test shall be performed in accordance with the applicable test requirements of MIL-DTL-25427. The flange shall be installed with MIL-DTL-25427/2 coupling half before test.

6. Interchangeability. The flange shall be interchangeable among manufacturers and shall be available to be mounted with MIL-DTL-25427/2 coupling half. This coupling flange configuration and dimensions are known to be similar to Parker coupling flange 44125, Eaton coupling flange 155-22, and Hydraulic International coupling flange MF. Suppliers seeking to be on QPL-25427 are responsible for interchangeability and shall perform the installation tests to ensure manufactured coupling flange is interchangeable with other supplier's coupling flange and vice versa.

7. Intended use. These mounting flanges are intended for use in military aircraft where the bulkhead coupling half is mounted on the aircraft through the flange delivering type II hydraulic fluid having a temperature range from -65 °F to +275 °F.

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CONCLUDING MATERIAL

Custodians:
Army-AV
Navy-AS
Air Force-99

Preparing activity:
Navy-AS

(Project 4730-2008-130)

Review activities:
Army-AT, MI
Navy-SA
Air Force-11
DLA-CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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